

DICTIONARY

```
In [3]: # Dictionary is a mutable data type in Python.  
# A python dictionary is a collection of key and value pairs separated by a colo  
# Keys must be unique in a dictionary, duplicate values are allowed.
```

```
In [4]: mydict = dict()  
mydict
```

```
Out[4]: {}
```

```
In [5]: mydict = {}  
mydict
```

```
Out[5]: {}
```

```
In [6]: mydict = {1:'one' , 2:'two' , 3:'three'} # dictionary with integer keys  
mydict
```

```
Out[6]: {1: 'one', 2: 'two', 3: 'three'}
```

```
In [7]: mydict = dict({1:'one' , 2:'two' , 3:'three'}) # Create dictionary using dict()  
mydict
```

```
Out[7]: {1: 'one', 2: 'two', 3: 'three'}
```

```
In [8]: mydict.keys()
```

```
Out[8]: dict_keys([1, 2, 3])
```

```
In [9]: mydict.values()
```

```
Out[9]: dict_values(['one', 'two', 'three'])
```

```
In [10]: mydict.items()
```

```
Out[10]: dict_items([(1, 'one'), (2, 'two'), (3, 'three')])
```

```
In [13]: mydict = {1:'one' , 2:'two' , 'A':{'Name':'asif' , 'Age':20}, 'B':('Bat' , 'ca')  
mydict
```

```
Out[13]: {1: 'one', 2: 'two', 'A': {'Name': 'asif', 'Age': 20}, 'B': ('Bat', 'ca')}
```

```
In [14]: keys = {'a','b','c','d'}  
mydict3 = dict.fromkeys(keys)  
mydict3
```

```
Out[14]: {'b': None, 'a': None, 'c': None, 'd': None}
```

```
In [16]: keys = {'a','b','c','d'}  
value = 10  
mydict3 = dict.fromkeys(keys,value)  
mydict3
```

Out[16]: {'b': 10, 'a': 10, 'c': 10, 'd': 10}

```
In [17]: keys = {'a' , 'b' , 'c' , 'd'}
value = [10,20,30]
mydict3 = dict.fromkeys(keys , value) # Create a dictionary from a sequence of
mydict3
```

Out[17]: {'b': [10, 20, 30], 'a': [10, 20, 30], 'c': [10, 20, 30], 'd': [10, 20, 30]}

```
In [18]: mydict1 = {'Name':'Asif' , 'ID': 74123 , 'DOB': 1991 , 'job' : 'Analyst'}
mydict1
```

Out[18]: {'Name': 'Asif', 'ID': 74123, 'DOB': 1991, 'job': 'Analyst'}

```
In [19]: mydict1['Name']
```

Out[19]: 'Asif'

```
In [21]: mydict1.get('Name')
```

Out[21]: 'Asif'

```
In [22]: mydict1['DOB'] = 1992 # Changing Dictionary Items
mydict1['Address'] = 'Delhi'
mydict1
```

Out[22]: {'Name': 'Asif',
'ID': 74123,
'DOB': 1992,
'job': 'Analyst',
'Address': 'Delhi'}

```
In [23]: dict1 = {'DOB':1995}
mydict1.update(dict1)
mydict1
```

Out[23]: {'Name': 'Asif',
'ID': 74123,
'DOB': 1995,
'job': 'Analyst',
'Address': 'Delhi'}

```
In [30]: mydict1.pop('job')
```

```
-----
KeyError                                Traceback (most recent call last)
Cell In[30], line 1
----> 1 mydict1.pop( )

KeyError: 'job'
```

```
In [31]: mydict1.popitem()
```

Out[31]: ('Address', 'Delhi')

```
In [32]: all(mydict1)
```

Out[32]: True

In [33]: `any(mydict1)`

Out[33]: True

In []: